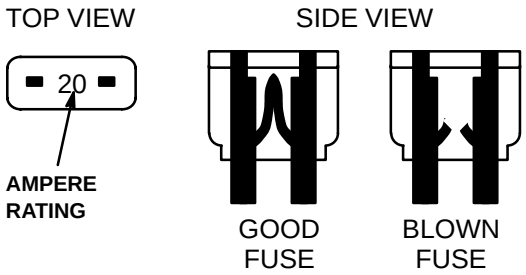
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, fuse link cartridges, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

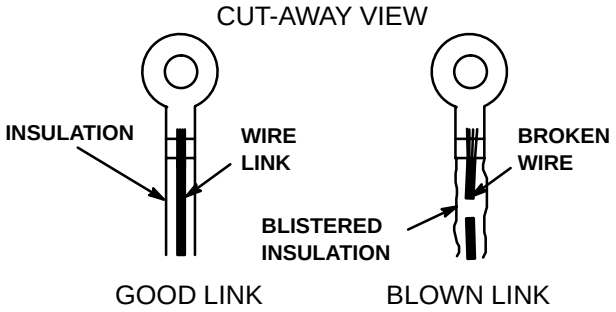


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

FUSIBLE LINK



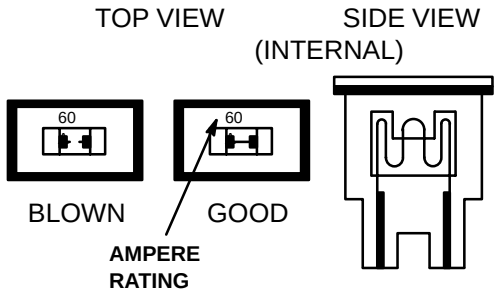
Fusible links are short lengths of wire that are smaller in diameter than the wires they are protecting. Fusible link wire is covered with a special thick, non-flammable insulation. An overload condition causes the insulation to blister. If the overload condition continues, the wire link will melt. To check a fusible link, look for blistered insulation. If the insulation is okay, pull lightly on the wire; If the fusible link stretches, the wire has melted.

When replacing fusible links, first cut the protected wire where it is connected to the fusible link. Then, tightly crimp or solder the new link to the protected wire.

Fusible links are often identified by color coding of the insulation, as shown here:

FUSIBLE LINK COLOR CODING	
WIRE LINK SIZE	**INSULATION COLOR**
20 GA	Blue
18 GA	Brown or Red
16 GA	Black or Orange
14 GA	Green
12 GA	Gray

FUSE LINK CARTRIDGE



Fuse link cartridges have a colored plastic housing with a clear “window” at the top. To check a fuse link cartridge, look at the fuse element through the clear “window.”

To replace a fuse link cartridge, pull it from the fuse box or panel. High ampere cartridges (80 A or more) are held in place by screws, which must be removed in order to replace the cartridge. Always replace a blown fuse link cartridge with a new one having the same ampere rating.

The ampere rating of a fuse link cartridge can also be determined by following the color code shown here:

FUSE LINK CARTRIDGE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
20	Light Blue
30	Pink
40	Green
50	Red
60	Yellow
80	Black
100	Dark Blue

CIRCUIT BREAKER

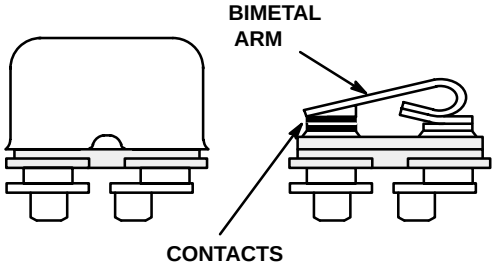
Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart).
They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other the arm bends, opening the contacts. Current flow is broken. A circuit breaker can be the cycling or non-cycling type.

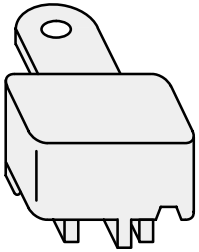
FUSE PANEL MOUNTED CYCLING TYPE

SIDE VIEW (EXTERNAL)

SIDE VIEW (INTERNAL)



IN-LINE MOUNTED CYCLING TYPE

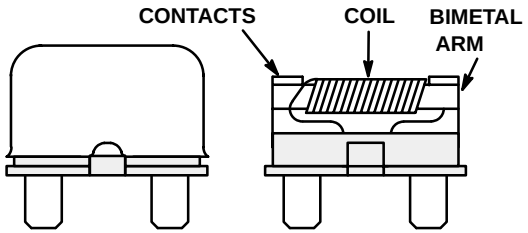


In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

FUSE PANEL MOUNTED NON-CYCLING TYPE

SIDE VIEW (EXTERNAL)

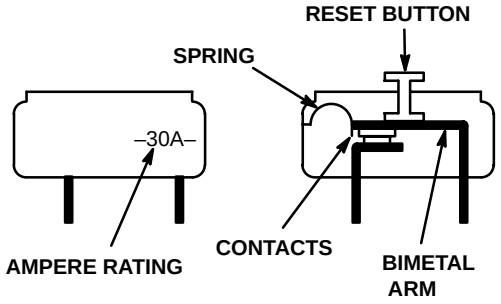
SIDE VIEW (INTERNAL)



FUSE PANEL MOUNTED MANUAL RESET TYPE

SIDE VIEW (EXTERNAL)

SIDE VIEW (INTERNAL)

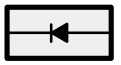


Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

DIODE



Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

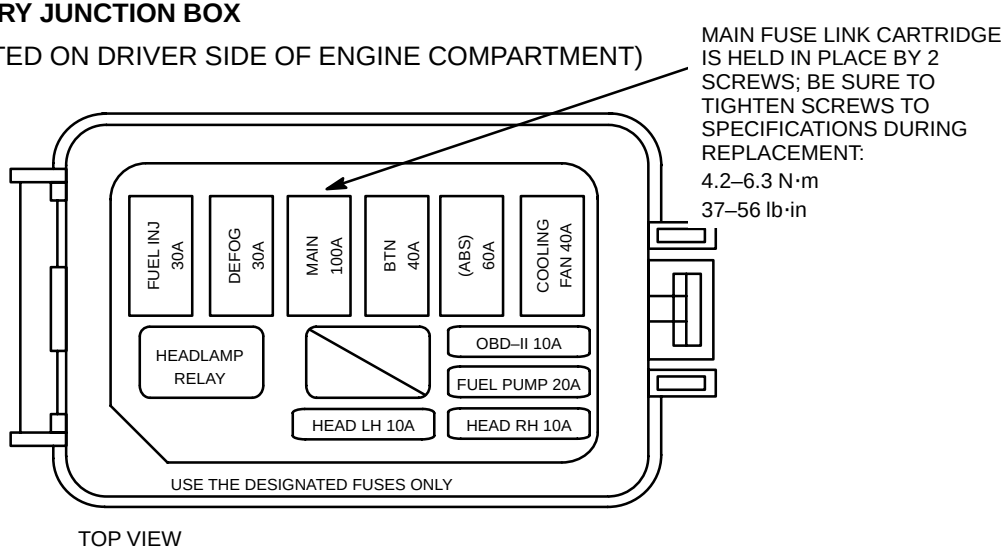
11-3 FUSE PANEL/CIRCUIT PROTECTION

FUSE LINK CARTRIDGE	AMPS	CIRCUITS PROTECTED
FUEL INJ.	30	AIR BAGS, ENGINE CONTROLS, GENERATOR
DEFOG.	30	REAR WINDOW DEFROST
MAIN	100	OVERALL CIRCUIT PROTECTION
BTN	40	HAZARD STOP, DOOR LOCK, TAIL, ROOM AND HORN FUSES OF THE CENTRAL JUNCTION BOX
ABS	60	ABS MAIN RELAY
COOLING FAN	40	CONSTANT CONTROL RELAY MODULE
FUSE	**AMPS**	**CIRCUITS PROTECTED**
OBD-II	10	DATA LINK CONNECTOR (DLC), INSTRUMENT CLUSTER
FUEL PUMP	20	ENGINE CONTROLS
HEAD RH	10	HEADLAMPS
HEAD LH	10	HEADLAMPS

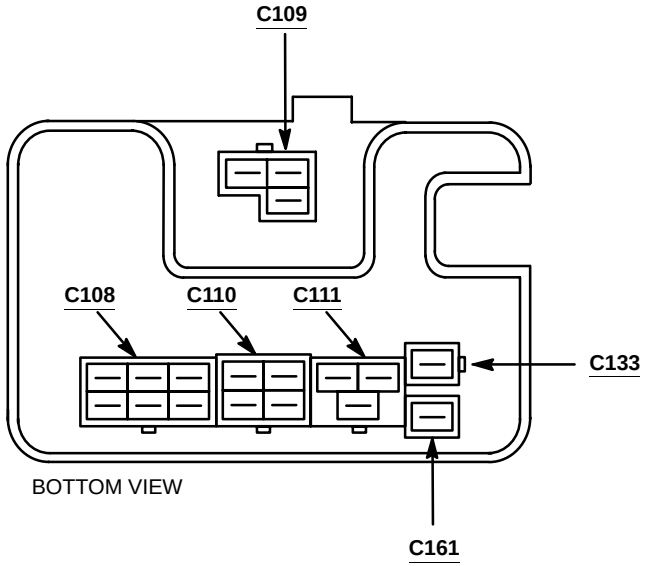
AMPERE RATING	HOUSING COLOR
BLADE TYPE FUSE	
10	Red
15	Blue
20	Yellow
30	Green
FUSE LINK CARTRIDGE	
20	Light Blue
30	Pink
40	Green
50	Red
60	Yellow
80	Black
100	Dark Blue

BATTERY JUNCTION BOX

(LOCATED ON DRIVER SIDE OF ENGINE COMPARTMENT)



TOP VIEW

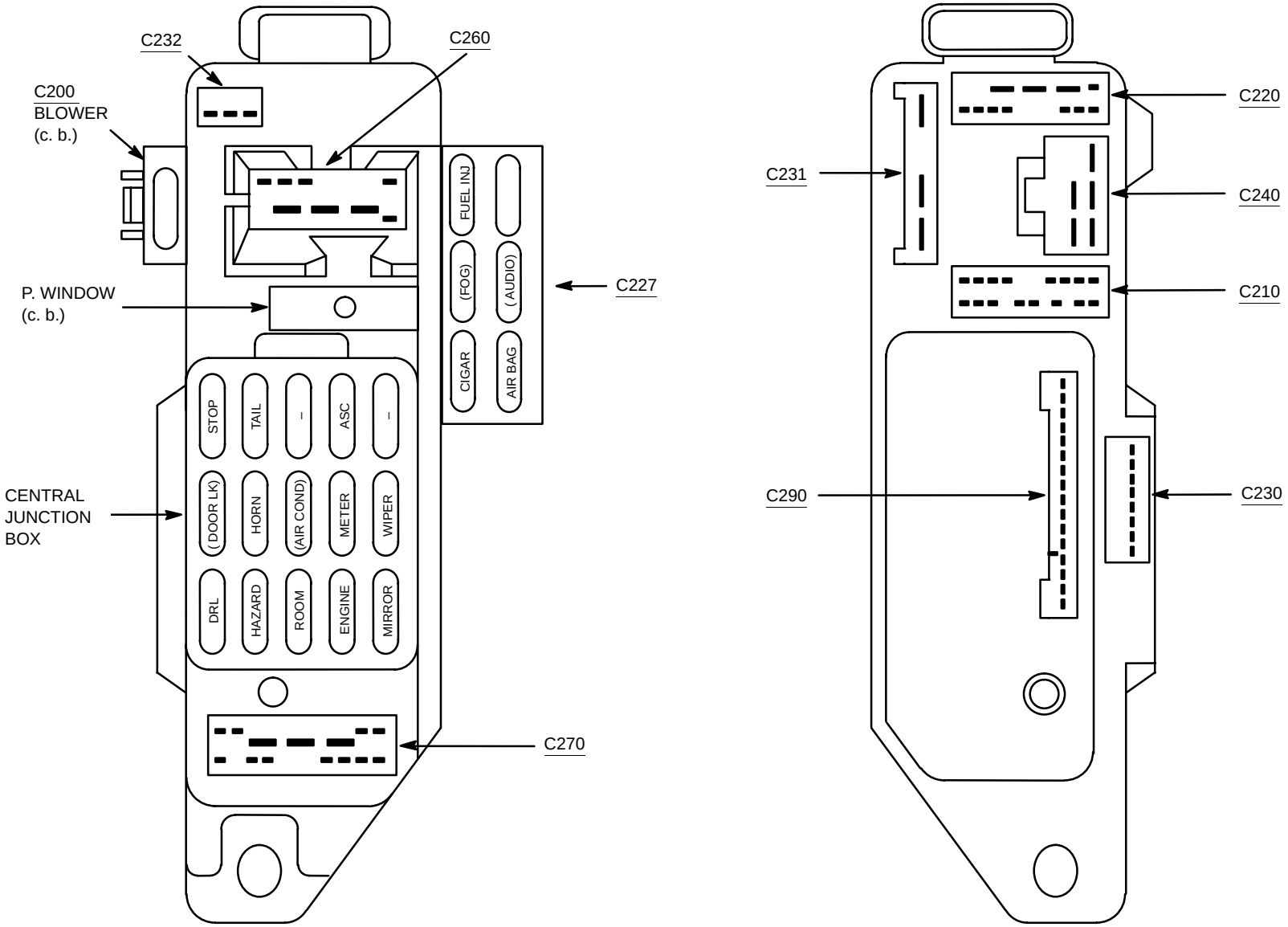


BOTTOM VIEW

CENTRAL JUNCTION BOX

(LOCATED ABOVE THE LEFT SIDE KICK PANEL)

SEDAN



11-5 FUSE PANEL/CIRCUIT PROTECTION

FUSE	AMPS	CIRCUITS PROTECTED
ASC	10	SPEED CONTROL
BLOWER (C.B.)	30	BLOWER MOTOR RELAY
CIGAR	20	CIGAR LIGHTER
ENGINE	15	ENGINE CONTROLS, TR SENSOR
FOG	10	FOG LAMPS
HAZARD	15	ELECTRONIC FLASHER, POWERTRAIN CONTROL MODULE (PCM)
AIR COND.	15	A/C-HEATER, ANTI-LOCK BRAKES
HORN	15	HORN
DOOR LOCK	30	POWER DOOR LOCKS
METER	10	REVERSING LAMPS, ENGINE COOLANT LEVEL SWITCH, INSTRUMENT CLUSTER, REAR WINDOW DEFROST, SHIFT LOCK, WARNING CHIME, TURN SIGNAL SWITCH
MIRROR	5	EXTERIOR REAR VIEW MIRRORS, RADIO, REMOTE KEYLESS ENTRY (RKE)
P. WINDOW (C.B.)	30	POWER WINDOWS
AUDIO	–	(NOT USED)
ROOM	10	REMOTE ANTI-THEFT PERSONALITY (RAP) MODULE, RADIO, SHIFT LOCK, COURTESY LAMPS, STARTING SYSTEM, WARNING CHIME
DRL	–	(NOT USED)
STOP	15	STOPLAMPS, BRAKE PRESSURE SWITCH
TAIL	15	PARK LAMP RELAY
WIPER	20	WIPER/WASHER, BLOWER MOTOR RELAY
FUEL INJECTOR	10	HO2S, EVAPORATIVE EMISSION PURGE FLOW SENSOR
AIR BAG	10	JOINT CONNECTOR #12, AIR BAG DIAGNOSTIC MONITOR

SEDAN

AMPERE RATING	HOUSING COLOR
BLADE TYPE FUSE	
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

POWER DISTRIBUTION

The Generator, Battery and Starter Motor are connected through the 100 Amp Main Fuse in the Battery Junction Box. Other circuits originate from this Main Fuse and are protected by additional fuses in the Battery Junction Box and Central Junction Box.

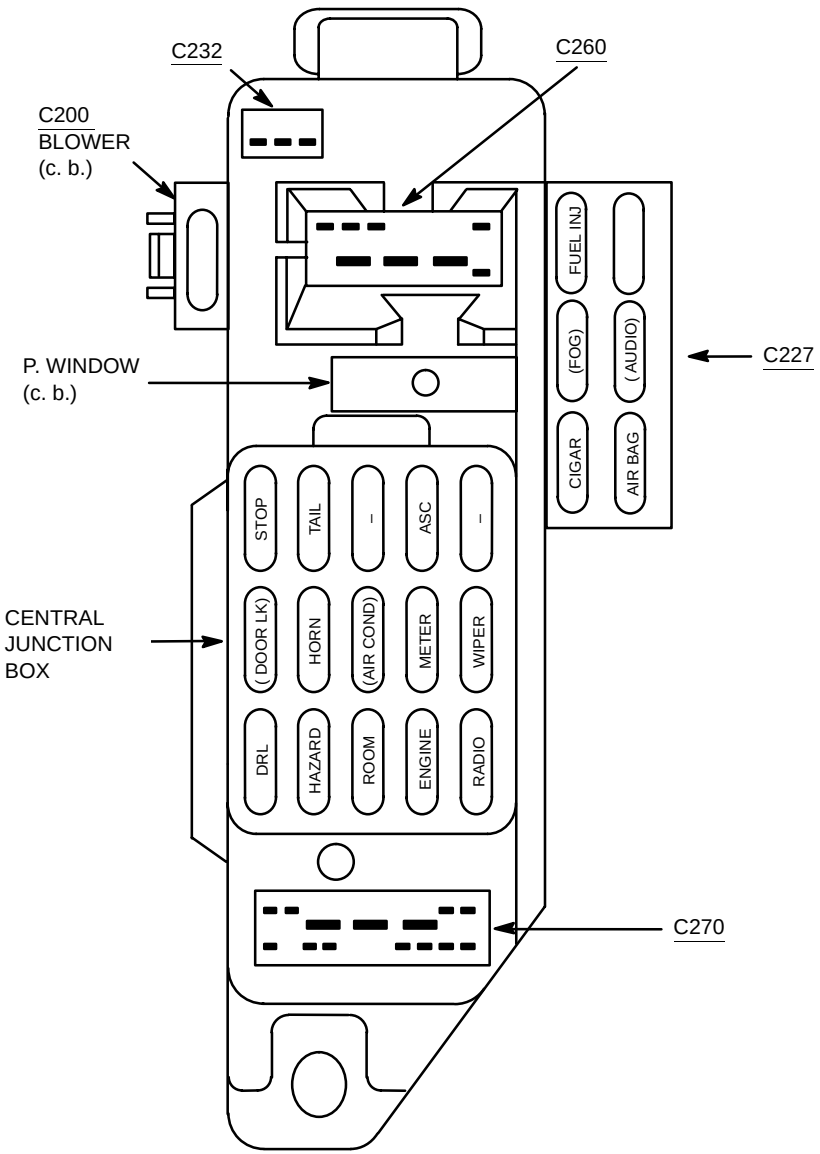
The fuses in the Battery Junction Box are powered at all times; as are the Blower, Door Lock, Fog, Hazard, Horn, Radio, Room, Stop and Tail Fuses in the Central Junction Box, the Ignition Switch and the Main Light Switch. The other fuses are powered through the Ignition Switch or the Main Light Switch.

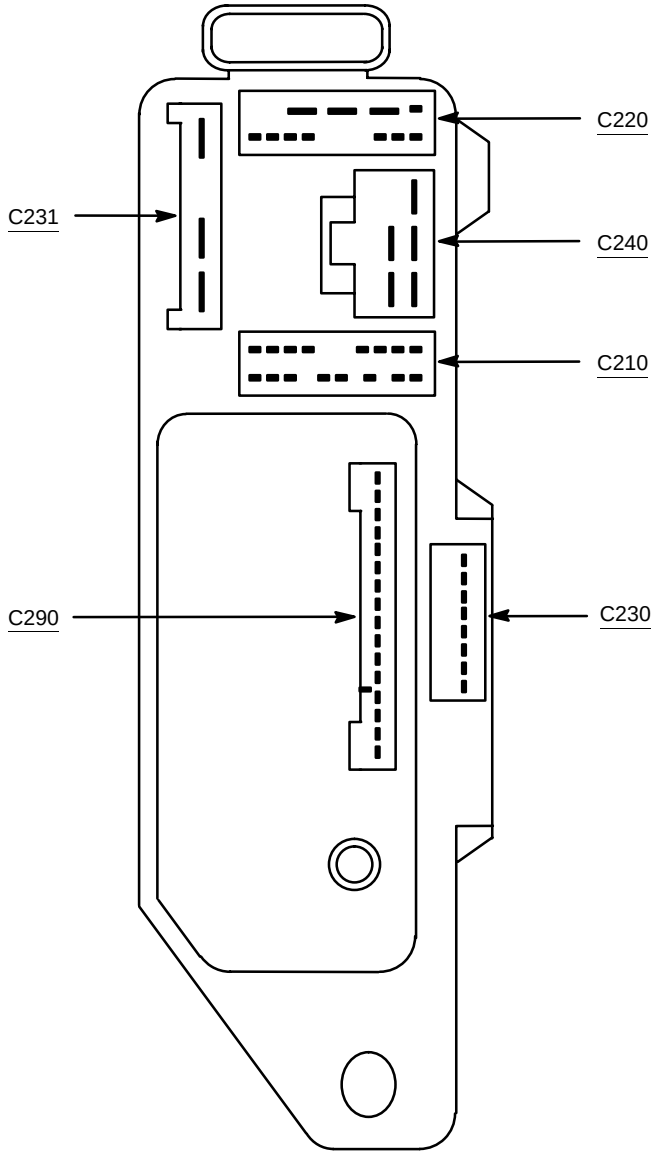
NOTE: The Daytime Running Lamps (DRL) Circuit is protected by an in-line 10A fuse that is part of the DRL Resistor.

CENTRAL JUNCTION BOX

(LOCATED ABOVE THE LEFT SIDE KICK PANEL)

COUPE





11-7 FUSE PANEL/CIRCUIT PROTECTION

FUSE	AMPS	CIRCUITS PROTECTED
ASC	10	SPEED CONTROL
BLOWER (C. B.)	30	BLOWER MOTOR RELAY
CIGAR	20	CIGAR LIGHTER
ENGINE	15	ELECTRONIC AUTOMATIC TRANSAXLE, IGNITION SYSTEM, CONSTANT CONTROL RELAY MODULE
FOG	10	FOG LAMPS, DAYTIME RUNNING LAMPS (DRL)
HAZARD	15	ELECTRONIC FLASHER, POWERTRAIN CONTROL MODULE (PCM)
AIR COND.	15	A/C-HEATER, ANTI-LOCK BRAKES
HORN	15	HORN, SHIFT LOCK
DOOR LOCK	30	POWER DOOR LOCKS
METER	10	REVERSING LAMPS, CHECK FUEL CAP INDICATOR, INSTRUMENT CLUSTER, REAR WINDOW DEFROST, SHIFT LOCK, WARNING CHIME, TURN SIGNAL SWITCH
RADIO	5	EXTERIOR REAR VIEW MIRRORS, RADIO, REMOTE KEYLESS ENTRY (RKE)
P. WINDOW (C. B.)	30	POWER WINDOWS
AIR BAG	10	JOINT CONNECTOR #12, AIR BAG DIAGNOSTIC MONITOR
ROOM	10	RAP SYSTEM, RADIO, SHIFT LOCK, COURTESY LAMPS, STARTING SYSTEM, WARNING CHIME, INSTRUMENT CLUSTER
FUEL INJECTOR	10	HO2S, EVAPORATIVE EMISSION PURGE FLOW SENSOR
DRL	10	DAYTIME RUNNING LAMPS (DRL)
STOP	20	STOPLAMPS, DEACTIVATOR SWITCH
TAIL	15	PARK LAMP RELAY
WIPER	20	WIPER/WASHER, BLOWER MOTOR RELAY
ROOF OPENING	15	ROOF OPENING PANEL
AUDIO	15	PREMIUM SOUND AMPLIFIER, CD CHANGER

COUPE

AMPERE RATING	HOUSING COLOR
BLADE TYPE FUSE	
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

POWER DISTRIBUTION

The Generator, Battery and Starter Motor are connected through the 100 Amp Main Fuse in the Battery Junction Box. Other circuits originate from this Main Fuse and are protected by additional fuses in the Battery Junction Box and Central Junction Box.

The fuses in the Battery Junction Box are powered at all times; as are the Blower, Door Lock, Fog, Hazard, Horn, Radio, Room, Stop and Tail Fuses in the Central Junction Box, the Ignition Switch and the Main Light Switch. The other fuses are powered through the Ignition Switch or the Main Light Switch.

NOTE: The Daytime Running Lamps (DRL) Circuit is protected by an in-line 10A fuse that is part of the DRL Resistor.